time, set out a rigorous theoretical framework for understanding electricity, magnetism, and their intimate relationship.[1] Maxwell developed these equations by carefully studying the work of the English physicist Michael Faraday, who in the early 1800s had carried out tens of thousands of experiments that exposed hitherto unknown features of electricity and magnetism. Faraday's key breakthrough was the concept of the *field*. Later expanded on by Maxwell and many others, this concept has had an enormous influence on the development of physics during the last two centuries, and underlies many of the little mysteries we encounter in everyday life. When you go through airport security, how is it that a machine that doesn't touch you can determine whether you're carrying metallic objects? When you have an MRI, how is it that a device that remains outside your body can take a detailed picture of your insides? When you look at a compass, how is it that the needle swings around and points north even though nothing seems to nudge it? The familiar answer to the last question invokes the earth's magnetic field, and the concept of magnetic fields helps to explain the previous two examples as well.

I've never seen a better way to get a visceral sense of a magnetic field than the elementary school demonstration in which iron filings are sprinkled in the vicinity of a bar magnet. After a little shaking, the iron filings align themselves in an orderly pattern of arcs that begin at the magnet's north pole and swing up and around, to end at the magnet's south pole, as in Figure 3.1. The pattern traced by the iron filings is direct evidence that the magnet creates an invisible something that permeates the space around it—a something that can, for example, exert a force on shards of metal. The invisible something is the *magnetic field* and, to our intuition, it resembles a mist or essence that can fill a region of space and thereby exert a force beyond the physical extent of the magnet itself. A magnetic field provides a magnet what an army provides a dictator and what auditors provide the IRS: influence beyond their physical boundaries, which allows force to be exerted out in the "field." That is why a magnetic field is also called a force field.